03/04/2025

CODE-A

EDITION 2025 EDITION

Aakash

Medical|IIT-JEE|Foundations

Corporate Office: AESL, 3rd Floor, Incuspaze Campus-2, Plot-13,
Sector-18, Udyog Vihar, Gurugram, Haryana-122015

720

AIM – 720

(*Advanced* INTENSIVE *Mastery for* 720)

MM : 720

PST-4

Time : 180 Mins.

Answers

1. (3)	37. (1)	73. (3)	109. (4)	145. (3)
2. (2)	38. (1)	74. (2)	110. (2)	146. (1)
3. (4)	39. (3)	75. (1)	111. (1)	147. (4)
4. (3)	40. (1)	76. (2)	112. (1)	148. (2)
5. (3)	41. (1)	77. (2)	113. (3)	149. (4)
6. (3)	42. (3)	78. (2)	114. (2)	150. (2)
7. (3)	43. (2)	79. (4)	115. (2)	151. (2)
8. (3)	44. (1)	80. (2)	116. (3)	152. (1)
9. (4)	45. (3)	81. (3)	117. (2)	153. (2)
10. (3)	46. (2)	82. (3)	118. (4)	154. (2)
11. (2)	47. (2)	83. (3)	119. (2)	155. (3)
12. (3)	48. (1)	84. (2)	120. (1)	156. (4)
13. (2)	49. (4)	85. (2)	121. (4)	157. (3)
14. (3)	50. (3)	86. (1)	122. (4)	158. (1)
15. (1)	51. (3)	87. (2)	123. (2)	159. (3)
16. (4)	52. (1)	88. (3)	124. (2)	160. (3)
17. (2)	53. (3)	89. (2)	125. (1)	161. (1)
18. (2)	54. (1)	90. (4)	126. (3)	162. (2)
19. (2)	55. (4)	91. (3)	127. (3)	163. (2)
20. (4)	56. (4)	92. (1)	128. (4)	164. (2)
21. (1)	57. (3)	93. (2)	129. (2)	165. (2)
22. (2)	58. (2)	94. (2)	130. (1)	166. (2)
23. (1)	59. (3)	95. (1)	131. (2)	167. (3)
24. (2)	60. (3)	96. (2)	132. (3)	168. (2)
25. (1)	61. (4)	97. (3)	133. (1)	169. (4)
26. (1)	62. (3)	98. (1)	134. (2)	170. (2)
27. (3)	63. (3)	99. (2)	135. (2)	171. (3)
28. (1)	64. (2)	100. (3)	136. (3)	172. (4)
29. (4)	65. (2)	101. (3)	137. (1)	173. (4)
30. (1)	66. (2)	102. (2)	138. (2)	174. (2)
31. (3)	67. (1)	103. (2)	139. (4)	175. (3)
32. (1)	68. (4)	104. (3)	140. (4)	176. (1)
33. (2)	69. (1)	105. (2)	141. (1)	177. (1)
34. (3)	70. (3)	106. (2)	142. (3)	178. (4)
35. (1)	71. (2)	107. (3)	143. (3)	179. (2)
36. (4)	72. (2)	108. (3)	144. (2)	180. (1)

Hints and Solutions

PHYSICS

(1) Answer : (3)
Solution:
$T.E. = \frac{-GMm}{2r}$
⇒ Change in total energy = $\frac{-GMm}{2\frac{4}{3}\times r} + \frac{GMm}{2r}$
$\Rightarrow \Delta E = \frac{GMm}{2r}\left[\frac{-3}{4}+1\right] = \frac{GMm}{8r}$

(2) Answer : (2)
Solution:
$g = \frac{GM}{r} = \frac{G}{r}\times\rho\times\frac{4}{3}\pi r^3$
$\Rightarrow g = G\times\frac{4}{3}\times\pi\times\rho\times r^2$
$V_e = \sqrt{\frac{2GM}{r}} = \sqrt{\frac{2}{r}\times gr^2} = \sqrt{2gr}$
$\frac{V_{\text{earth}}}{V_{\text{mars}}} = \sqrt{\alpha\times\beta} = \sqrt{\alpha\beta}$

(3) Answer : (4)
Solution:
$\vec{F}_{net} = \frac{d\vec{P}}{dt}$
$\vec{F}_{net} = 0$
$\frac{d\vec{P}}{dt} = 0$
$\vec{P}$ = constant
$\vec{v}$ = constant

(4) Answer : (3)
Solution:
$$\vec{A}\times\vec{B} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & -2 & 6 \\ 1 & -2 & 1 \end{vmatrix}$$
$= \hat{i}\left(-2+12\right) - \hat{j}\left(1-6\right) + \hat{k}\left(-2+2\right)$
$= 10\hat{i} + 5\hat{j}$

(5) Answer : (3)
Solution:
Location of centre of mass depends on the distribution of mass and $m_1r_1 = m_2r_2$, where r_1 and r_2 are distance of centre of mass from m_1 and m_2 respectively.

(6) Answer : (3)
Solution:
$\tau = I\,\alpha = I\times\frac{d\omega}{dt}$
$= I\times 2(1-2t) = 2I(1-2t)$

(7) Answer : (3)
Solution:
$I = ma^2 + m\times(a\cos 60^\circ)^2$
$= ma^2 + \frac{ma^2}{4} = \frac{5ma^2}{4}$

(8) Answer : (3)
Solution:
$\omega_i = \frac{2\times\pi\times 33}{60}, \alpha = \frac{\Delta\omega}{\Delta t}$
$\Rightarrow \alpha = \frac{0-\frac{2\pi\times 33}{60}}{20}$
$\Rightarrow \alpha = \frac{2\pi\times 33}{60\times 20} = -\frac{11\pi}{200}$ rad/s^2

(9) Answer : (4)
Solution:
$\frac{Gm}{x^2} = \frac{G\times 9m}{(R–x)^2} \Rightarrow \left(\frac{R-x}{x}\right)^2 = 9$
$\Rightarrow \frac{R\times x}{x} = 3 \Rightarrow R–x = 3x$
$\Rightarrow R = 4x \Rightarrow x = \frac{R}{4}$
$\Rightarrow V = \frac{-Gm}{\frac{R}{4}} - \frac{G\times 9m}{3\frac{R}{4}} = \frac{-4Gm}{R}[1+3]$
$= \frac{-16Gm}{R}$

(10) Answer : (3)
Solution:
$I_g = \frac{-F_g}{m} = \frac{3}{60\times 10^{-3}}$
$= \frac{3\times 10^3}{60}$ = 50 N/kg

(11) Answer : (2)
Solution:
for, $r \le R, g \propto r \rightarrow$ straight line variation
for, $r > R, g \propto \frac{1}{r^2}$

(12) Answer : (3)
Solution:
A liquid will wet the solid surface if angle of contact is acute i.e. less than 90°.

(13) Answer : (2)
Solution:
As the ball is dropped, the velocity will first increase and then it would attain terminal velocity at which net force will be balanced.

(14) Answer : (3)
Solution:
Kinetic energy per unit volume = $\frac{\frac{1}{2}mv^2}{V}$
$K' = \frac{1}{2}\rho v^2$
from equation of continuity:
$A_1v_1 = A_2v_2$
$\because A_P < A_Q \Rightarrow v_P > v_Q$
Hence, $K'_P > K'_Q$

(15) Answer : (1)
Solution:
Venturi meter works on Bernoulli's principle.

(16) Answer : (4)
Solution:
The angular momentum of the planet about sun remains constant while speed, angular speed and kinetic energy varies.

(17) Answer : (2)
Solution:
$g' = g\left[1-\frac{d}{R}\right]$ and $d = \frac{R}{2}$
$\Rightarrow g' = g\left[1-\frac{R}{2R}\right] = \frac{g}{2}$
⇒ weight = $\frac{250}{2}$ = 125 N

(18) Answer : (2)
Solution:
$B = \frac{-\Delta P}{\frac{\Delta V}{V}} = \frac{\Delta P}{\frac{\Delta \rho}{\rho}} \Rightarrow \frac{\Delta \rho}{\rho} = \frac{\Delta P}{B}$

$\Rightarrow 1 \times 10^{-3} = \frac{\Delta P}{2 \times 10^{9}} \Rightarrow \Delta P = 2 \times 10^{9} \times 10^{-3}$

= 2 × 10^6 N/m^2

(19) Answer : (2)
Solution:
Stress its mid point

= $\frac{\text{Weight}}{\text{Area}} = \frac{\frac{m}{L} \times \frac{L}{2} \times g}{A} = \frac{mg}{2A}$

$= \frac{g}{2A} \times \rho \times A \times L = \frac{1}{2} \times \rho \times g \times L$

(20) Answer : (4)
Solution:
Young's modulus is the property of material.

(21) Answer : (1)
Solution:
Elastic potential energy
= $\frac{1}{2}$ × Load × extension
= $\frac{1}{2} \times Mg \times l = \frac{Mgl}{2}$

(22) Answer : (2)
Solution:
$L_1 - L = \frac{M_0 \times g \times L}{A \times Y} \Rightarrow M_0 = AY\left(\frac{L_1}{L} - 1\right) \times \frac{1}{g}$

(23) Answer : (1)
Solution:
L = constant = $I \times \omega = I' \times \frac{3\omega}{4}$

$\Rightarrow I' = \frac{4I}{3} \Rightarrow \Delta I = \frac{4I}{3} - I = \frac{I}{3}$

(24) Answer : (2)
Solution:
$\vec{\tau} = \vec{r} \times \vec{F} = -2\hat{k} \times 3\hat{j} = 6\hat{i}$ N m

(25) Answer : (1)
Solution:
$\tau = I\alpha \Rightarrow 100 = 300 \times \alpha \Rightarrow \alpha = \frac{1}{3}$ rad/s^2

$\omega = \omega_0 + \alpha t \Rightarrow \omega = \frac{1}{3} \times 2 = \frac{2}{3}$ rad/s

(26) Answer : (1)
Solution:
Using conservation of energy, $mg\frac{L}{2}(1 - \cos\theta) = \frac{1}{2} \times I \times \omega^2$

$\Rightarrow mg\frac{L}{2}(1 - \cos\theta) = \frac{1}{2} \times \frac{mL^2}{3} \times \omega^2$

$\Rightarrow \sqrt{\frac{6g}{L}} \sin\left(\frac{\theta}{2}\right)$

(27) Answer : (3)
Solution:
As no net external force is acting
so V_{com} = 0.
$\Rightarrow m \times v = 2m \times v' \Rightarrow v' = \frac{v}{2}$

(28) Answer : (1)
Solution:
$\frac{GmM}{R^2} = m\omega^2 R \Rightarrow w^2 = \frac{GM}{R^3}$

$\Rightarrow \omega^2 = \frac{G}{R^3} \times d \times \frac{4}{3}\pi R^3$

$\omega^2 = \frac{4Gd\pi}{3} \Rightarrow \frac{4\pi}{T^2} = \frac{4Gd}{3} \Rightarrow T^2 = \frac{3\pi}{Gd}$

$\Rightarrow T = \sqrt{\frac{3\pi}{Gd}}$

(29) Answer : (4)
Solution:

$\overrightarrow{E} = \frac{-\partial v}{\partial x}\hat{i} \quad \frac{-\partial v}{\partial y}\hat{j} - \frac{\partial v}{\partial z}\hat{k}$

$\overrightarrow{E} = \frac{-2K}{X^3}\hat{i} \Rightarrow \overrightarrow{E} = \frac{-2\times K}{8} = \frac{-K}{4}\hat{i}$

(30) Answer : (1)
Solution:

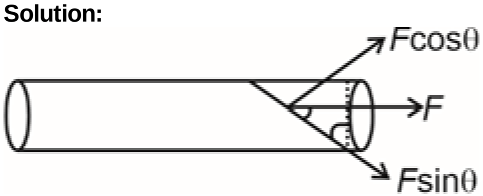
$T^2 \propto r^3 \Rightarrow \left(\frac{24}{T}\right)^2 = \left[\frac{7R_E}{2R_E}\right]^3$

$\Rightarrow \frac{24\times 24}{T^2} = \frac{49\times 7}{4\times 2}$

$T^2 = \frac{24\times 24\times 4\times 2}{49\times 7}$

$T = \frac{24\times 2}{7}\sqrt{\frac{2}{7}} = \frac{48}{7}\sqrt{\frac{2}{7}}\mathrm{h}$

(31) Answer : (3)
Solution:



$\tau = \frac{F\sin\theta}{\frac{A}{\cos\theta}} = \frac{F}{A}\sin\theta\cos\theta$

$= \frac{F}{2A} \times \sin\left(2\theta\right)$

(32) Answer : (1)
Solution:
Young's modulus is given by slope of stress strain curve.

$Y_A = \tan 45^\circ,\ Y_B = \tan 30^\circ$

$Y_B = \frac{Y_A}{\sqrt{3}}$

$\Rightarrow Y_A = \sqrt{3}\,Y_B$

(33) Answer : (2)
Solution:

$\frac{\text{Breaking strength}}{\text{Area}}$ = Breaking stress

$\Rightarrow \sigma = \frac{4\times 10^5}{\frac{\pi}{4}\times 3\times 3} = \frac{F}{\frac{\pi}{4}\times 1.5\times 1.5} \Rightarrow F = 1\times 10^5\mathrm{N}$

(34) Answer : (3)
Solution:
Apparent weight = Actual weight + Force due to surface tension
⇒ Apparent weight – Actual weight
= s × 2 × (l + t)
= 0.073 × 2 × [0.1 + 0.004]
= 0.015 N

(35) Answer : (1)
Solution:

$F = \Delta p \times A = 10^3 \times 10 \times 5 \times 10^{-3} = 50\ \mathrm{N}$

$v = \sqrt{2gh} = \sqrt{2\times 10\times 5}$ = 10 m/s

Volume flow rate = Av = 10^{-3} × 10 = 0.01 m^3/s

(36) **Answer :** (4)
Solution:
When two masses come closer, their gravitational potential energy decreases.
In case of a couple, linear momentum is constant but angular momentum is not constant.
Pressure is not a vector quantity.

(37) **Answer :** (1)
Solution:
$g' = \frac{g}{\left(1+\frac{h}{R}\right)^2} \Rightarrow \left(1+\frac{h}{R}\right)^2 = \frac{g}{\frac{g}{4}} \Rightarrow 1+\frac{h}{R} = 2$

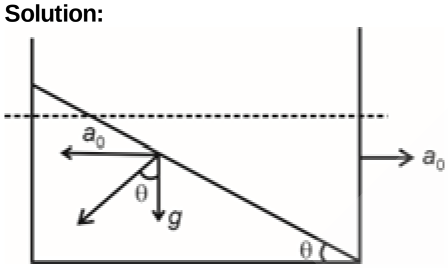
$\frac{h}{R} = 1 \Rightarrow h = R$

(38) **Answer :** (1)
Solution:
$F = \frac{Gm_1m_2}{r^2}$

$G = \frac{(MLT^{-2})L^2}{M^2}$

The correct dimensional formula for gravitational constant is $\left[M^{-1}L^3T^{-2}\right]$

(39) **Answer :** (3)
Solution:
$F_{up} = \rho V_{imm} g$
Since both the pieces are experiencing equal upthrust, hence they must have equal volume.

(40) **Answer :** (1)
Solution:


$\tan\theta = \frac{a_0}{g}$
or $\theta = \tan^{-1}\left(\frac{a_0}{g}\right)$
The free surface of liquid will be perpendicular to net acceleration as liquid at rest cannot sustain shear stress.

(41) **Answer :** (1)
Solution:
Force = $2 \times 2\pi \times r \times T = 4\pi rT$

(42) **Answer :** (3)
Solution:
$\Delta P = \frac{2T}{r} \Rightarrow \frac{2T}{r_1} = 5 \times \frac{2T}{r_2} \Rightarrow \frac{r_2}{r_1} = 5$

$\Rightarrow \frac{A_2}{A_1} = \frac{r_2^2}{r_1^2} = \frac{25}{1}$

(43) **Answer :** (2)
Solution:
According to Newton's law of viscosity,
$\frac{F}{A} = -\eta \times \frac{dv}{dy} \Rightarrow F = -\eta A \frac{dv}{dy}$

(44) **Answer :** (1)
Solution:
Max load = Max stress × Area of cross section
Since area of cross-section and maximum stress remains same hence maximum load it can sustain will remain same.

(45) **Answer :** (3)
Solution:

Balancing torque about fulcrum,

$10g\frac{l}{2} = mg\frac{l}{2} + 3g \times \frac{3l}{2}$

$5l = \frac{ml}{2} + \frac{9l}{2}$

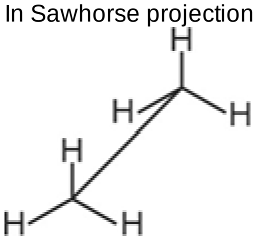
m = 1 kg

CHEMISTRY

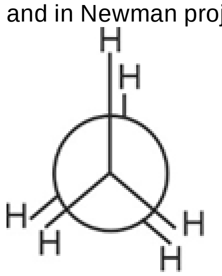
(46) **Answer :** (2)

Solution:

In Sawhorse projection



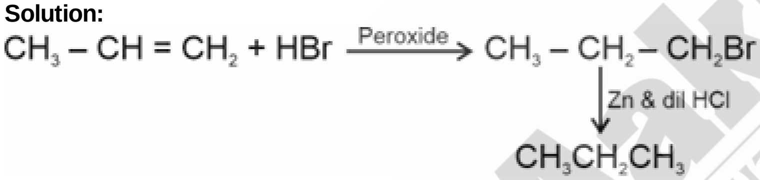
and in Newman projection



are eclipsed.

(47) **Answer :** (2)

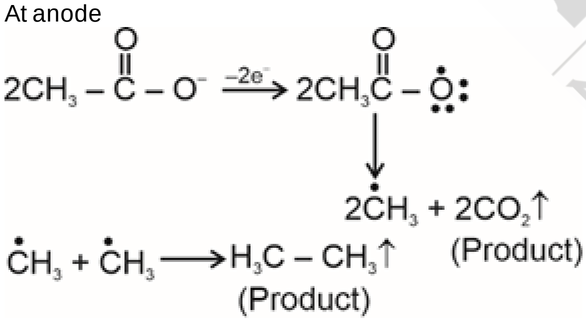
Solution:



(48) **Answer :** (1)

Solution:

At anode



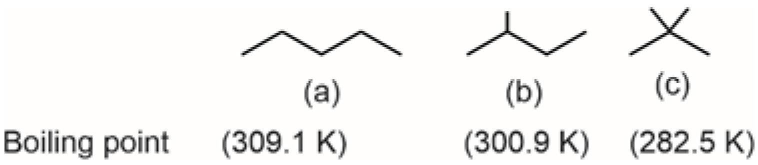
(49) **Answer :** (4)

Solution:

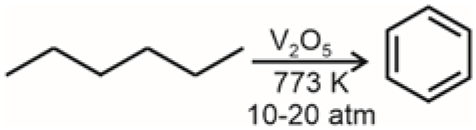
$$2\,\mathrm{CH_4} + \mathrm{O_2} \xrightarrow[100\ \mathrm{atm}]{\mathrm{Cu}\ /523\,\mathrm{K}/} 2\,\mathrm{CH_3\,OH}$$

(50) **Answer :** (3)

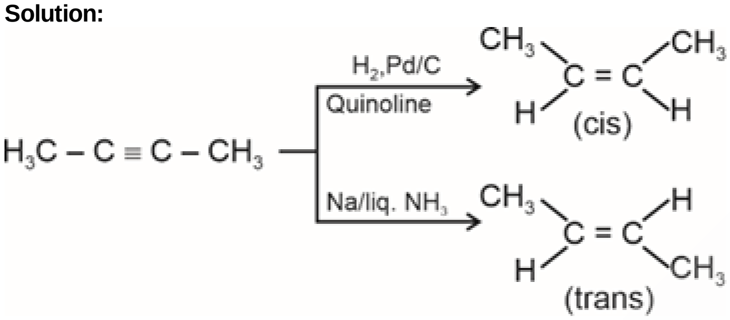
Solution:



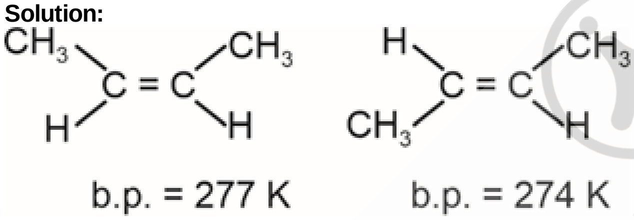
(51) Answer : (3)
Solution:



(52) Answer : (1)
Solution:

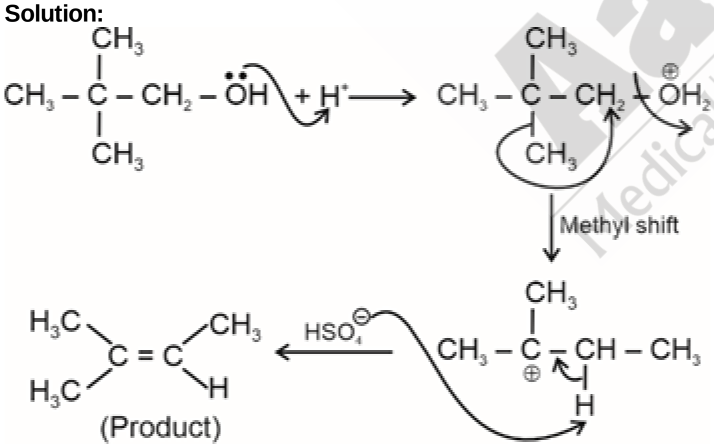


(53) Answer : (3)
Solution:



and trans isomer is having higher melting point

(54) Answer : (1)
Solution:

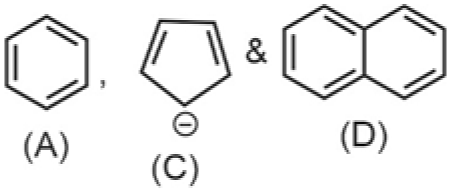


(55) Answer : (4)
Hint:
$\Delta H = \Delta U + \Delta n_g RT$
Solution:
• $N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$, $\Delta n_g = -2$
$\Delta H = \Delta U - 2RT$
• $PCl_5(g) \rightarrow PCl_3(g) + Cl_2(g)$, $\Delta n_g = 1$
$\Delta H = \Delta U + RT$
• $C(s) + O_2(g) \rightarrow CO_2(g)$, $\Delta n_g = 0$
$\Delta H = \Delta U$

• $CO(g)+\frac{1}{2}O_2(g)\rightarrow CO_2(g),\ \Delta n_g=\frac{-1}{2}$

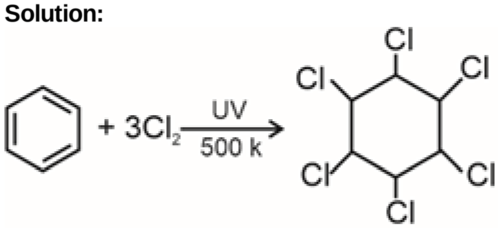
$\Delta H=\Delta U-\frac{1}{2}RT$

(56) **Answer :** (4)
Solution:



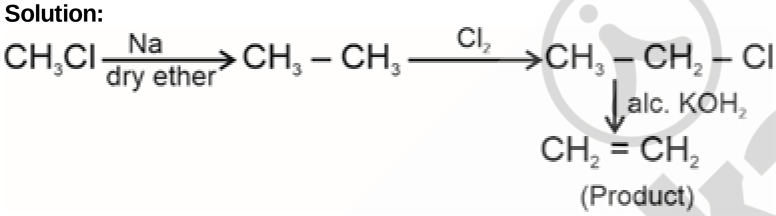
are aromatic compounds

(57) **Answer :** (3)
Solution:

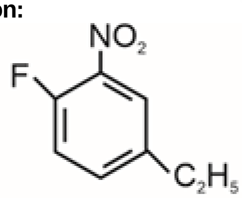


(Benzene hexachloride)
BHC

(58) **Answer :** (2)
Solution:



(59) **Answer :** (3)
Solution:



4-Ethyl-1-fluoro-2-nitrobenzene

(60) **Answer :** (3)
Solution:
More will be hyperconjugating structures more will be stability

$^{\bullet}C(CH_3)_3 > {}^{\bullet}CH(CH_3)_2\ {}^{\bullet}CH_2CH_3 > {}^{\bullet}CH_3$

(61) **Answer :** (4)
Solution:

Molar mass of AgBr = 188 g mol^{-1}
188 g AgBr contains 80 g bromine
0.08 g AgBr contains $\frac{80}{188}\times 0.08\ g$ bromine

% of bromine = $\frac{80\times 0.08\times 100}{188\times 0.1}=34.04\%$

(62) **Answer :** (3)
Solution:

(63) Answer : (3)
Solution:
Order –COOH > – SO_3H > –COOR > –CHO

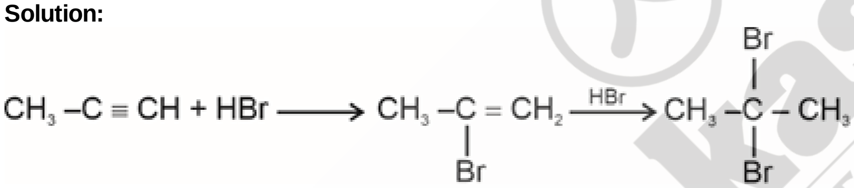
(64) Answer : (2)
Solution:
$6\,CN^- + Fe^{2+} \rightarrow [Fe\,(CN)_6]^{4-}$

$3[Fe\,(CN)_6]^{4-} + 4Fe^{3+} \xrightarrow{xH_2\,O} Fe_4\,[Fe\,(CN)_6]_3 \cdot xH_2\,O$

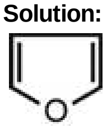
(65) Answer : (2)
Solution:
$CaC_2 + H_2O \rightarrow HC \equiv CH \uparrow + Ca\,(OH)_2$

(66) Answer : (2)
Solution:
12 g of C yields 44 g of CO_2
$\frac{12}{44} \times 0.099$ of C yields 0.099 g of CO_2
% of carbon = $\frac{12}{44} \times \frac{0.099}{0.126} \times 100 = 21.42\%$

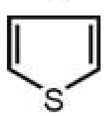
(67) Answer : (1)
Solution:



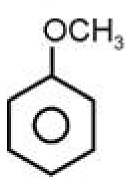
(68) Answer : (4)
Hint:
Methoxy benzene is known as Anisole.
Solution:



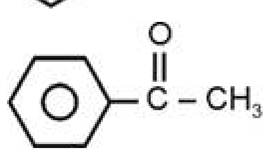
→ Furan



→ Thiophene



→ Anisole



→ Acetophenone

(69) Answer : (1)
Solution:
In adiabatic process, dq = 0
Using first law, dU = dq + dw

$\boxed{dU = dw}$

(70) Answer : (3)
Solution:
$dS = \frac{dq}{T}$
dU = dq + dw ; dU = 0 (as ΔT = 0)
dq = –dw
$dS = \frac{-(-PdV)}{T}$
$\Delta S = nR \ \ln\frac{V_2}{V_1}$
$\Delta S = 2.303 \ nR \ \log\frac{V_2}{V_1}$
$\Delta S = 2.303 \times \ 10 \times R \times \ \log\frac{10}{100}$
ΔS = –2.303 × 10 × R
ΔS = –23.03R

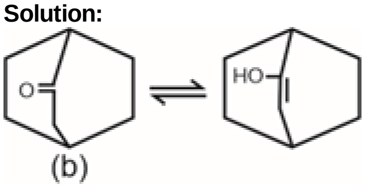
(71) Answer : (2)
Solution:
Pressure ⇒ Intensive property
Heat capacity = Extensive property
q & w are path function

(72) Answer : (2)
Solution:
$\Delta H = \Delta U + n_gRT$
$n_g = (5 - 3 - 2) = 0$
ΔG = ΔH – TΔS
ΔG = –10000 – 300(–30)
ΔG = –10000 + 9000
ΔG = –1000 J

(73) Answer : (3)
Solution:
For isothermal condition, ΔU = 0 and free expansion means ext P = 0
$w = -P_{ext}dV = 0$
Using first law of thermodynamics, $dU = dq + dw$
ΔU = q

(74) Answer : (2)
Solution:
ΔG = –ve, for a reaction to be called as spontaneous.

(75) Answer : (1)
Solution:
–OH has + M increases the electron density over the benzene ring, hence more reactive to electrophilic attack.

(76) Answer : (2)
Solution:


Only (b) can tautomerise to enol

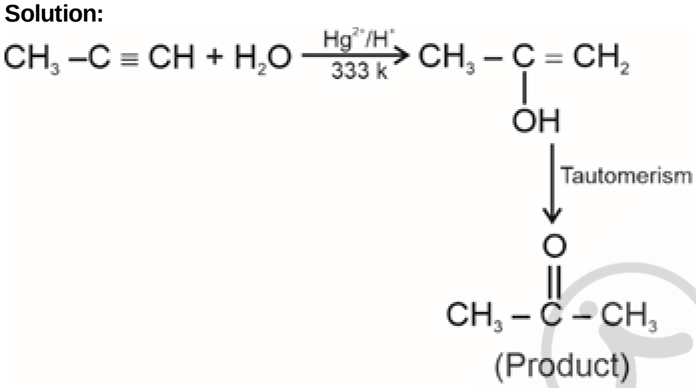
(77) Answer : (2)
Solution:
On reversing the reaction (ii) & add (i) with it, we get so heat of formation of SO_2 is (x – 2y) k cal
$S(s) + O_2(g) \rightarrow SO_2(g) + (x-2y)$
So, heat of formation of SO_2 is (2y – x) kcal

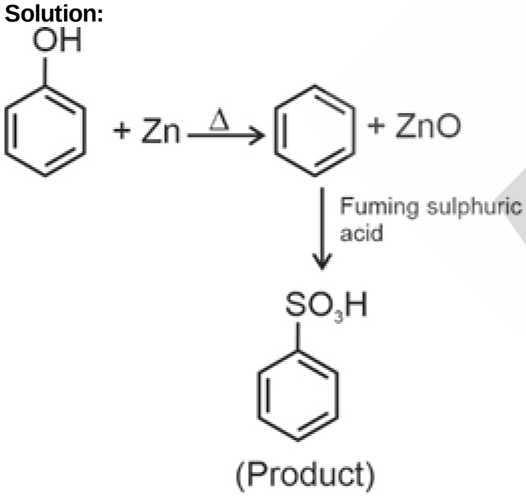
(78) Answer : (2)
Solution:
When a system is in equilibrium, the entropy is maximum and change in entropy is zero

(79) **Answer :** (4)
Solution:
$\frac{q_{rev}}{T} = \Delta S$, which is state function.

(80) **Answer :** (2)
Solution:
$2H_2(g) \rightarrow 4\,H(g)$; ΔH = +869.6 kJ
$H_2(g) \rightarrow 2\,H(g)$; $\Delta H = \frac{+869.6}{2}$ kJ = 434.8 kJ mol^{-1}

(81) **Answer :** (3)
Solution:
ΔU = q + w
q = +600 J (heat is absorbed)
w = –250 J (as done by the system)
So, ΔU = 600 – 250 = 350 J

(82) **Answer :** (3)
Solution:


(83) **Answer :** (3)
Solution:


(84) **Answer :** (2)
Solution:
w = 0.16 g, V_1 = 46 mL, 15°C = 273 + 15 = 288 K

$\frac{(P - P_1)\,V_1}{T_1} = \frac{P_2 V_2}{T_2}$

$\frac{(755 - 15)\,46}{288} = \frac{760}{273} \times V_2,\ V_2 = 42.6\ \text{mL}$

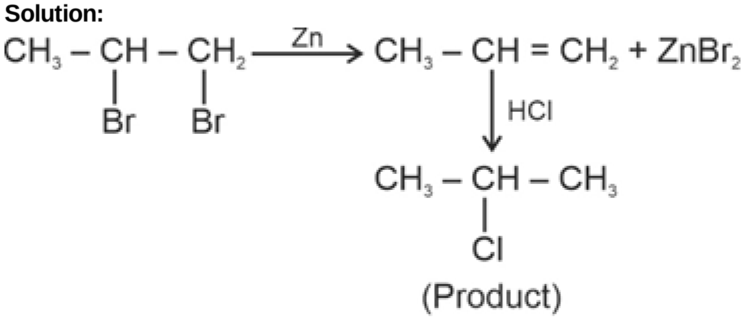
$\%N = \frac{28}{22400} \times \frac{V_2(\text{mL})}{w} \times 100$

$\%N = \frac{28}{22400} \times \frac{42.6}{0.16} \times 100 = 33.28\%$

(85) **Answer :** (2)
Solution:

Halogens are estimated using Carius method
Kjeldahl's method is not applicable to compounds containing nitrogen in nitro compounds, azo compounds and nitrogen present in ring

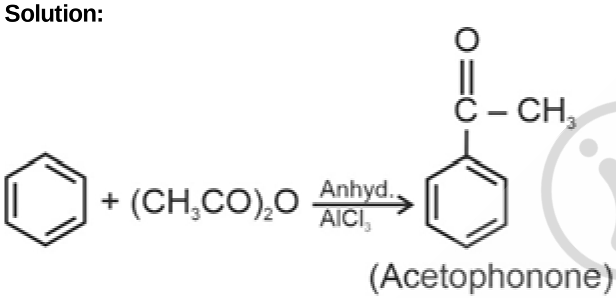
(86) Answer : (1)
Solution:



(87) Answer : (2)
Solution:

$$CH_3-CH=CH-CH_3 \xrightarrow[\text{(ii) Zn /}H_2O]{\text{(i) }O_3} \underset{\text{(Product)}}{2\,CH_3\,CHO}$$

(88) Answer : (3)
Solution:



(89) Answer : (2)
Solution:
Halogenation of alkanes follows free radical mechanism & rate of reaction follows : $F_2 > Cl_2 > Br_2 > I_2$.

(90) Answer : (4)
Solution:
Column chromatography and thin layer chromatography are based on the principle of differential adsorption.

BOTANY

(91) Answer : (3)
Solution:
Tomato is a C_3 plant while maize is a C_4 plant. In former, photorespiration occurs while in latter, photorespiration does not occur.

(92) Answer : (1)
Solution:
Pyruvic acid gets converted into Acetyl CoA in the mitochondrial matrix by pyruvate dehydrogenase that requires Mg^{2+} ions.

(93) Answer : (2)
Solution:
In T. W. Engelmann experiment, aerobic bacteria were used to detect the site of O_2 evolution.

(94) Answer : (2)
Solution:
Auxins and ethylene promote flowering in pineapple.

(95) Answer : (1)

Solution:
Centrioles are responsible for aster formation in animal cells and they duplicate during S phase of interphase and during interkinesis or intra-meiotic interphase.

(96) Answer : (2)
Solution:
Rate of photosynthesis is measured by the O_2 released. Higher the O_2 released, higher will be the graph of action spectrum of photosynthesis.

(97) Answer : (3)
Solution:
During the conversion of glyceraldehyde-3-phosphate into 1, 3-bisphosphoglyceric acid, one molecule of NADH + H^+ is formed.

(98) Answer : (1)
Solution:
Apical bud secretes auxin which inhibits the growth of lateral buds. If the apical bud is removed (decapitation), the lateral buds sprout i.e., start growing.

(99) Answer : (2)
Solution:
Anaerobic respiration occurs in cytoplasm.

(100) Answer : (3)
Solution:
Cells that do not divide further, exit G_1 phase to enter an inactive stage called quiescent stage (G_0) of the cell cycle.

(101) Answer : (3)
Solution:
S or synthesis phase marks the period during which DNA synthesis or replication and centriole duplication takes place.

(102) Answer : (2)
Solution:
The Calvin cycle proceeds in three stages.
(i) Carboxylation: CO_2 combines with ribulose-1,5-bisphosphate.(b)
(ii) Reduction: Carbohydrate is formed at the expense of the photochemically made ATP and NADPH.(c)
(iii) Regeneration: CO_2 acceptor ribulose-1,5-bisphosphate is formed again so that the cycle continues.(a)

(103) Answer : (2)
Solution:
TCA cycle starts with the condensation of acetyl group with oxaloacetic acid (4C) and water to yield citric acid (6C). There are three points in the cycle where NAD^+ is reduced to NADH + H^+. The continued oxidation of acetyl CoA via the TCA cycle requires the continued replenishment of oxaloacetic acid.

(104) Answer : (3)
Solution:
ABA induces dormancy of buds, seeds and storage organs.

(105) Answer : (2)
Solution:

Leptotene	Condensation and coiling of chromatin fibres
Zygotene	Two homologous chromosomes form a pair
Diakinesis	Terminalization of chiasmata
Diplotene	Dictyotene stage

(106) Answer : (2)
Solution:
Prophase of mitosis can be characterised by:
- Untangling of chromosomes
- Condensation of chromosomal material
- Spireme stage

(107) Answer : (3)
Solution:
Cells at the end of prophase, when viewed under the microscope, do not show Golgi complexes endoplasmic reticulum, nucleolus and the nuclear envelope.

(108) Answer : (3)
Solution:
Zeatin is a cytokinin i.e. growth promoter.

(109) Answer : (4)
Solution:
Kinetin and N^6-furfurylamino purine are adenine derivatives.

(110) Answer : (2)
Solution:
Along with auxins, extract of meristematic tissues, yeast extract, coconut DNA and kinetin can be used to proliferate a mass of undifferentiated cells in a nutrient medium.

(111) Answer : (1)
Solution:
During oxidation within a cell, all the energy contained in respiratory substrates is not released free into the cell or in a single step. It is released in a series of slow step-wise reactions controlled by enzymes and it is trapped as chemical energy in the form of ATP.

(112) Answer : (1)
Solution:
Both auxin and cytokinin promote cell division which shows their synergistic effect on cell division.

(113) Answer : (3)
Solution:
Ethylene inhibits longitudinal growth but stimulates transverse growth of seedlings.

(114) Answer : (2)
Solution:
The first natural cytokinin was isolated from unripe maize grains or kernels known as zeatin.

(115) Answer : (2)
Solution:
Auxins such as IAA and IBA in diluted form are used to produce parthenocarpic or seedless fruits.

(116) Answer : (3)
Solution:
Protoplasmic modification occurs at phase of differentiation.

(117) Answer : (2)
Solution:
Water is one of the raw materials utilized for the process of photosynthesis. Photosynthetic process utilises less than 1% of the water absorbed by a plant, hence it is rarely a limiting factor in photosynthesis.

(118) Answer : (4)
Solution:
Within the chloroplasts, the membranes the are sites for light reaction, while the chemosynthetic pathway occurs in the stroma.

(119) Answer : (2)
Solution:
The NADP reductase enzyme is located on the stroma side of the membrane. Protons are necessary for the reduction of $NADP^+$ to $NADPH + H^+$ and these protons are removed from the stroma.

(120) Answer : (1)
Solution:
M phase is the most dramatic period of the cell cycle, involving a major reorganisation of virtually all components of the cell.

(121) Answer : (4)
Solution:
The complete disintegration of the nuclear envelope marks the start of the second phase of mitosis i.e., Metaphase. Movement of centrosome to opposite poles occurs in prophase.

(122) Answer : (4)
Solution:
At telophase stage, nuclear envelope develops around the chromosome clusters at each pole, forming two daughter nuclei.

(123) Answer : (2)
Solution:

White blood cell is an animal cell and it divides its cytoplasm by furrow formation in the plasma membrane.

(124) Answer : (2)
Solution:
The stroma lamellae lacks PS II as well as NADP reductase enzyme.

(125) Answer : (1)
Solution:
C_3 plants are similar to C_4 plants as both undergo Calvin cycle. Calvin cycle occurs in all photosynthetic plants.

(126) Answer : (3)
Solution:
During decarboxylation reaction in bundle sheath cell of C_4 plant, the C_4 acid is converted into C_3 acid, that gets transported from bundle sheath cell to mesophyll cell. This acid is regenerated into phosphoenolpyruvate (PEP).

(127) Answer : (3)
Solution:
Alcohol fermentation results in the release of CO_2 along with ethanol while lactic acid fermentation releases lactic acid only.

(128) Answer : (4)
Solution:
Oxidative phosphorylation is formation of ATP by energy released from electrons removed during substrate oxidation.
Substrate level phosphorylation is formation of ATP by transfer of phosphate group from a substrate to ADP.

(129) Answer : (2)
Solution:
Complex II (succinate dehydrogenase), converts succinate into fumarate.

(130) Answer : (1)
Solution:
GA_3 is used to speed up the malting process in brewing industry. Gibberellins stimulate elongation of stem and other aerial parts. However, it has no effect on roots.

(131) Answer : (2)
Solution:
Increase if protoplasm is difficult to measure directly. Therefore, growth can be measured by various parameters which are more or less proportional to it.

(132) Answer : (3)
Solution:
The correct sequence of cell cycle is G_1, S, G_2 and M.

(133) Answer : (1)
Solution:
Animals can show equational division in haploid cells. Pachytene is long lived than zygotene.

(134) Answer : (2)
Solution:
Crossing over is mediated by recombinase. This process occurs in pachytene.

(135) Answer : (2)
Solution:
The bivalent chromosome aligns on the equatorial plate during metaphase I.

ZOOLOGY

(136) Answer : (3)
Solution:

Respiratory Gas	Atmospheric Air	Alveoli	Blood (Deoxygenated)	Blood (Oxygenated)	Tissues
O_2	159	104	40	95	40
CO_2	0.3	40	45	40	45

Both pulmonary artery and systemic vein carry deoxygenated blood.

(137) Answer : (1)
Solution:

Blood Group	Antigens on RBCs	Antibodies in Plasma	Donor's Group
A	A	Anti-B	A, O
B	B	anti-A	B, O
AB	A, B	nil	AB, A, B, O
O	nil	anti-A, B	O

Donor's blood group should not have Rh antigens on his RBCs surface, if recipient has Rh –ve blood group. Otherwise, recipient's body will form antibodies against Rh antigen.

(138) Answer : (2)
Solution:
Emphysema is the disorder in which alveolar walls are damaged due to which respiratory surface is decreased. One of the major causes of this is cigarette smoking.
Asthma is a difficulty in breathing causing wheezing due to inflammation of bronchi and bronchioles.

(139) Answer : (4)
Solution:
During swallowing of food, glottis can be covered by epiglottis to prevent the entry of food into larynx.

(140) Answer : (4)
Solution:
In the alveoli, where there is high pO_2, low pCO_2, lesser H^+ concentration and lower temperature, the factors are favourable for the formation of oxyhaemoglobin.
In tissues, where low pO_2, high pCO_2, high H^+ concentration and higher temperature exist, the conditions are favourable for dissociation of oxygen from oxyhaemoglobin.

(141) Answer : (1)
Solution:
The tubular epithelial cells in different segments of nephron perform reabsorption either by active or passive mechanisms.
Eg., substances like glucose, amino acids, Na^+, *etc*, in the filtrate are reabsorbed actively whereas the nitrogenous wastes are reabsorbed by passive transport. Reabsorption of water also occurs passively in the initial segments of nephrons.

(142) Answer : (3)
Solution:
Brush border epithelium is present in PCT for reabsorption of various electrolytes and water.
The glomerular capillary blood pressure causes filtration of blood through 3 layers, *i.e.*, the endothelium of glomerular blood vessels, the epithelium of Bowman's capsule and a basement substance between these two layers.

(143) Answer : (3)
Solution:
Blood is medium of transport for O_2 and CO_2.

%	Gas	Carried by
97	O_2	RBCs in blood
3	O_2	Plasma in dissolved state
20-25	CO_2	RBCs in blood
70	CO_2	Plasma as bicarbonate
7	CO_2	Plasma in dissolved state

(144) Answer : (2)
Solution:
Kidneys are reddish brown, bean shaped structure situated between the levels of last thoracic and third lumbar vertebrae close to the dorsal inner wall of the abdominal cavity. Last thoracic vertebra is T_{12} and third lumbar vertebra is represented by L_3.

(145) Answer : (3)
Solution:
The nodal musculature of human heart has the ability to generate action potentials without any external stimuli, *i.e.*, it is auto-excitable.

Normal activities of the heart are regulated intrinsically, *i.e.*, autoregulated by specialised muscles (nodal tissue), hence the heart is called myogenic.

(146) Answer : (1)
Solution:
Lungs possess alveoli which do not collapse even after forceful expiration because of residual volume which is the volume of air remaining in the lungs even after a forceful expiration.

(147) Answer : (4)
Solution:
In majority of nephrons in human body, the loop of Henle is too short and extends only very little into the medulla. Such nephrons are called cortical nephrons.
In some nephrons, the loop of Henle is very long and runs deep into the medulla, these are called juxtamedullary nephrons.
Efferent arteriole emerging from glomerulus forms a fine capillary network around the renal tubule called the peritubular capillaries.

(148) Answer : (2)
Solution:
Respiratory rhythm centre is located in medulla oblongata whereas pneumotaxic centre is located in pons region of the brain.
Neural signals from pneumotaxic centre reduce the duration of inspiration and moderate the functions of respiratory rhythm centre.

(149) Answer : (4)
Solution:
Larynx is a cartilaginous box which helps in sound production and hence called the sound box. During swallowing, glottis can be covered by a thin elastic cartilaginous flap called epiglottis to prevent the entry of food into the larynx.
Alveoli are not supported by cartilage.

(150) Answer : (2)
Solution:
Mode of respiration:

Adult *Rana tigrina*	Lungs; (Pulmonary respiration)
Pheretima	Moist skin (cuticle); (Cutaneous respiration)
Prawn	Gills; (Branchial respiration)
Human	Lungs; (Pulmonary respiration)

(151) Answer : (2)
Solution:
Maximum reabsorption of water and electrolytes occurs in PCT (part of nephron lined by simple cuboidal brush border epithelium).
Descending limb of Henle's loop concentrates the filtrate whereas ascending limb dilutes the filtrate.
Conditional reabsorption of Na^{+} and water takes place in DCT.

(152) Answer : (1)
Solution:
Lymphocytes make up 20-25% of the total leucocytes. They are of two major types-B and T. Both B and T lymphocytes are responsible for immune responses of the body.
Lymphocytes are agranulocytes. Monocytes and macrophages are phagocytic cells. Basophils secrete histamine, serotonin and heparin.

(153) Answer : (2)
Solution:
In fishes, the heart pumps out deoxygenated blood which is oxygenated by the gills and supplied to the body parts from where deoxygenated blood is returned to the heart.
In amphibians, the left atrium receives oxygenated blood. In birds, double circulation takes place. In mammals, right atrium receives the deoxygenated blood from vena cava.

(154) Answer : (2)
Solution:
In hepatic portal system, the hepatic portal vein carries blood from intestine to the liver before it is delivered to the systemic circulation.

(155) Answer : (3)
Solution:

In human, the systemic circulation starts with the pumping of oxygenated blood by the left ventricle to the aorta which is carried to all the body parts and the deoxygenated blood from body parts is collected by the veins and returned to the right atrium. Left atrium is included under pulmonary circulation.

(156) Answer : (4)
Solution:
In humans, about 70 mL of blood is pumped out by each ventricle during a cardiac cycle and it is called stroke volume.
Volume of blood pumped out by each ventricle of heart per minute is called cardiac output.
Heart rate × stroke volume = cardiac output

(157) Answer : (3)
Solution:
In ECG, a patient is hooked up to a monitoring machine that shows voltage traces on screen and makes the sound "pip… pip… pip… peeeee" as the patient goes into cardiac arrest.
Heart failure is not the same as cardiac arrest (when the heart stops beating) or a heart attack (when the heart muscle is suddenly damaged by an inadequate blood supply). Heart failure means state of heart when enough blood is not pumped out to meet the needs of the body.

(158) Answer : (1)
Solution:
Neural signals through the sympathetic nerves, release of adrenal medullary hormones and strenuous exercise and workout increases the cardiac output whereas neural signals through parasympathetic nerves decrease the cardiac output.

(159) Answer : (3)
Solution:
In an ECG, the QRS complex represents the depolarisation of the ventricles, which initiates the ventricular contraction just after Q wave. The P-wave represents the electrical excitation (depolarisation) of the atria. The T-wave represents the return of ventricles from excited to normal state (repolarisation). The end of the T-wave marks the end of ventricular systole.

(160) Answer : (3)
Solution:
The renal tubule begins with a double walled cup-like structure called Bowman's capsule.
Glomerulus along with Bowman's capsule forms renal corpuscle. PCT, DCT and Henle's loop are parts of renal tubule.

(161) Answer : (1)
Solution:
Before haemodialysis, the dialysing fluid has the same composition as that of plasma except the nitrogenous wastes (urea) so that urea can easily move out from blood of patient into the dialysing fluid, clearing the blood.
Urea is the chief nitrogenous waste produced by mammals. Proteins, minerals and glucose all are present in plasma as well as in dialysing fluid.

(162) Answer : (2)
Solution:

(a)	*Amphioxus*	–	Protonephridia/Flame cells
(b)	*Pheretima*	–	Nephridia
(c)	Prawn	–	Green glands
(d)	*Periplaneta*	–	Malpighian tubules

(163) Answer : (2)
Solution:
In an adult human under normal physiological conditions:
· On an average, 25-30 gm of urea is excreted out per day.
· 1.0 to 1.5 L of urine is excreted out per day.
· Average breathing rate is 12-16 times min^{-1}

(164) Answer : (2)
Solution:
Heparin is an anticoagulant released by basophils (granulocytes) which make up the 0.5 to 1% of the total WBCs.
RBCs are devoid of nucleus and are formed in red bone marrow of adults.

(165) Answer : (2)
Solution:
A signal is initiated by the stretching of the urinary bladder as it gets filled with urine. In response, the stretch receptors on the walls of the bladder send signal to the CNS. The CNS passes on motor messages to initiate the contraction of smooth muscles of the bladder and simultaneous relaxation of the urethral sphincter causing the release of urine. The process of release of urine is called micturition.

(166) Answer : (2)
Solution:
Glomerulus is a tuft of capillaries formed by the afferent arteriole – a fine branch of renal artery. The efferent arteriole emerging from the glomerulus forms a fine capillary network around the renal tubule called the peritubular capillaries. A minute vessel of this network runs parallel to the Henle's loop forming a 'U-shaped' vasa recta.

(167) Answer : (3)
Solution:
Haemoglobin is a red coloured iron containing pigment present in the RBCs. O_2 can bind with haemoglobin in a reversible manner to form oxyhaemoglobin.
Each haemoglobin molecule can carry a maximum of four molecules of O_2.

(168) Answer : (2)
Solution:
Every 100 mL of oxygenated blood can deliver around 5 mL of O_2 to the tissues and 100 mL of deoxygenated blood delivers around 4 mL of CO_2 to the alveoli.

(169) Answer : (4)
Solution:
A represents squamous epithelium of alveolar wall,
B represents basement substance,
C represents endothelium of blood capillaries,
D represents red blood cells in capillary

(170) Answer : (2)
Solution:
'A' here represents carbonic anhydrase which is present in RBCs and small amounts of it is present in blood plasma.

(171) Answer : (3)
Solution:
Terrestrial adaptation necessitated the production of lesser toxic nitrogenous wastes like urea and uric acid for conservation of water.
Ammonia, the most toxic nitrogenous waste, is readily soluble, is generally excreted by diffusion across body surface.

(172) Answer : (4)
Solution:
The epithelial cells of Bowman's capsule called podocytes are arranged in an intricate manner so as to leave some minute spaces called filtration slits or slit pores.

(173) Answer : (4)
Solution:
The condition erythroblastosis foetalis can be avoided by administering Anti-Rh antibodies to the Rh –ve mother immediately after the delivery of the first Rh +ve child.

(174) Answer : (2)
Solution:
Lymph does not carry large proteins. It contains immune cells such as lymphocytes. Lymphatic system collects lymph and drains it back to the major veins. Lymph has same mineral distribution as that of plasma.

(175) Answer : (3)
Solution:
During ventricular systole, the pressure inside the ventricles rises up to a significant level. The chordae tendinae prevents the valves from inverting into the atria which could lead to regurgitation of blood.

(176) Answer : (1)
Solution:
Endothelium is the inner lining of blood vessels.
Both arteries and veins have endothelium.
Veins carry blood towards the heart whereas arteries carry blood away from the heart. Lumen of artery is narrower as compared to veins which leads to high pressure of blood. Valves are present in veins.

(177) Answer : (1)
Solution:
ADH is secreted from hypothalamus and is released from posterior pituitary. It has vasoconstrictory effects on blood vessels.
Angiotensin II being a powerful vasoconstrictor increases GFR.

(178) Answer : (4)
Solution:

The part starting with the external nostrils up to the terminal bronchioles constitute the conducting part whereas the alveoli and their ducts form the respiratory or exchange part of the respiratory system.
The anatomical set up of lungs in thorax is such that any change in the volume of the thoracic cavity will be reflected in the lung (pulmonary cavity)

(179) Answer : (2)
Solution:
IC – Total volume of air a person can inspire after a normal expiration;
IC = TV + IRV
FRC – Volume of air that will remain in the lungs after a normal expiration;
FRC = ERV + RV
TLC – Total volume of air accommodated in the lungs at the end of a forced inspiration;
TLC = RV + ERV + TV + IRV or VC + RV
VC – Maximum volume of air a person can breathe in after a forced expiration.
VC = ERV + TV + IRV

(180) Answer : (1)
Solution:
O_2 and CO_2 are exchanged at alveoli by simple diffusion mainly based on pressure/concentration gradient. Solubility of the gases as well as the thickness of the membranes involved in diffusion are some important factors that can affect the rate of diffusion.
The reactivity of gases (O_2 and CO_2) does not affect their diffusion.